Odoo Migration Cookbook

Version 19 to Version 20

COMMUNITY EDITION AND ENTERPRISE EDITION · FIRST EDITION · SEPTEMBER 2026

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This cookbook lists the code replacements required when moving custom modules from Odoo 19.0 to Odoo 20.0. It covers **Odoo Community** and **Odoo Enterprise**. Each entry is a 19 → 20 change: a model, field, method, view xpath, QWeb directive, HTTP type, or JavaScript API that 19 accepted and 20 does not.

The replacements are taken from the official Odoo **19.0** and **20.0** source trees (`odoo/odoo` Community, including `base` and the standard addons, and the matching Enterprise addons).

Applies to

Mark	Meaning
`CE + EE`	Same change in Community 20 and Enterprise 20
`CE`	Community 20 only, or a Community model that Enterprise does not use
`EE`	Enterprise 20 only (Payroll, Budget, Field Service, Sign, Enterprise Helpdesk, Enterprise Accounting extras)

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Confirm each replacement against the Odoo 20.0 Community or Enterprise source you deploy.

About the authors

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1. Security

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CE + EE

Odoo 20 loads `ir.access.csv`. It does not load `ir.model.access.csv` plus XML `ir.rule` records. XML `<record model="ir.rule">` raises `KeyError: 'ir.rule'`.

Odoo 19

```
id,name,model_id:id,group_id:id,perm_read,perm_write,perm_create,perm_unlink
access_lock,lock,model_my_lock,base.group_user,1,1,1,0
```

Odoo 20 — `security/ir.access.csv`

```
id,name,model_id,group_id/id,operation,domain
access_my_lock,my.lock,my.lock,base.group_user,cru,
```

Column	Odoo 20	
`model_id`	Technical model name (`sale.order`). Not `module.model_x`.	
`group_id/id`	Group XML id. Empty = global rule.	
`operation`	`crud`, `cru`, `cr`, … — not `rc`, and not `perm_read` columns	
`domain`	Record-rule domain, or empty	

Further 19 → 20 rules on this file:

- `model_id/id` from 19 is not `model_id`. The 20 column needs the technical

name.

- Domains `[(1, '=', 1)]` and `[('1', '=', 1)]` fail validation. Use an empty

domain.

- A domain on a non-stored field raises `Field has no SQL representation` . Use a

stored field or a stored path.

- Company domain `('company_id', 'in', company_ids + [False])` fails. Use:

```
['|', ('company_id', '=', False), ('company_id', 'in', company_ids)]
```

- Custom `res.groups` XML must be listed in the manifest **before**

`ir.access.csv` if the CSV references those groups.

2. ORM and Python

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CE + **EE** unless marked otherwise.

Odoo 19	Odoo 20
`_sql_constraints = [('name', 'sql', 'msg')]`	`_name = models.Constraint('sql', 'msg')`
`self.fields_view_get(...)`	`self._get_view(...)`
`self.check_access_rights('read')`	`self.check_access('read')`
`self.check_access_rights('read', raise_exception=False)`	`self.has_access('read')` (bool)
`self.user_has_groups('sales_team.group_sale_manager')`	`self.env.user.has_groups('sales_team.group_sale_manager')`
`self._context`	`self.env.context`
`self.env.user.company_id`	`self.env.company`
`ir.config_parameter.get_param` / `set_param`	`get_int` / `get_str` / `get_bool` `set_str` / `set_bool` / `set_float`
`pytz`	`datetime.UTC` and `zoneinfo`
window action `toggle_active`	`action_archive` / `action_unarchive`
write `archive=True`	write `active=False`
`mail.thread.cc`	`mail.thread`

```
# 19
_sql_constraints = [
    ('code_uniq', 'unique(code, company_id)', 'Code must be unique per
company.'),
]
# 20
_code_uniq = models.Constraint(
    'unique(code, company_id)',
    'Code must be unique per company.',
)
```

```
# 19
self.check_access_rights('read')
# 20
self.check_access('read')
self.has_access('read')
```

```
# 20 — selection_add requires ondelete
state = fields.Selection(selection_add=[
    ('extra', 'Extra'),
], ondelete={'extra': 'set default'})
```

`_name` listed in `_inherit` requires that parent model to exist in the registry. **CE** a module that `_name = 'helpdesk.sla'` and also `_inherit = 'helpdesk.sla'` fails unless **EE** `helpdesk` is installed. **CE-only** graphs use `_name` only. **EE** graphs that load Enterprise Helpdesk keep `_inherit` .

`mail.thread.cc` is removed. A model that inherits `rating.mixin` already inherits `mail.thread` . Listing both on 20 raises an MRO error.

```
# 20
_inherit = ['portal.mixin', 'mail.activity.mixin', 'rating.mixin']
```

Two Many2many fields on the same model to the same comodel, without `relation=` , collide on 20. Set `relation=` on at least one.

A 20 compute must assign every record. A 20 compute must not `return` a value dict. Stored and non-stored fields cannot share one compute method.

`method=True` on a computed field is invalid. `tracking=` on a related field requires `mail.thread`. Related Selection must not re-declare `selection` / `default` / `tracking`. `placeholder=` is XML-only.

A required SQL column without `default=` raises `NOT NULL` on create. `required=True` on a new Many2one cannot add `NOT NULL` when the table already has rows.

`self.env['some.model']` is a recordset. `_inherit` is on `self.env.registry['some.model']`.

COPY, WRITE ORDER, AND TRANSLATIONS

A Char field named `name` now defaults to `copy=mark_as_copy('name')`. Duplicating a record appends "(copy)" to that value. Override `copy=` when the 19 behaviour must stay a plain copy.

```
from odoo.tools.translate import mark_as_copy
name = fields.Char(copy=mark_as_copy('name'))
code = fields.Char(copy=False)
```

`create()` and `write()` run field inverses in `(field.write_sequence, field definition order)`. Custom inverses that assumed 19 order can break. Set `write_sequence=` when two inverses must run in a fixed sequence.

```
partner_id = fields.Many2one('res.partner', write_sequence=10)
```

Reading a One2many or Many2many returns **only records the user can access**. Code that counted or mapped the full relational cache of forbidden ids will see a shorter recordset.

Module translations are stored in **that module's PO**. 20 no longer dumps them into `base`. Export and import translations per module.

WEBSITE ON THE ENVIRONMENT

CE + EE when `website` is installed. 19 model code often used `request.website`. 20 Environment exposes `env.website` from

`context['website_id']` . Prefer `self.env.website` in models. `request` is not bound in report parsers and several ORM contexts (Chapter 4).

```
# 20
website = self.env.website
```

3. Search and domains

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CE + EE

Odoo 19	Odoo 20
`_name_search`	`_rec_names_search` and/or `_search_display_name`
`from odoo.osv import expression` / `expression.AND` / `expression.OR`	`from odoo.fields import Domain` / `Domain.AND` / `Domain.OR`
`from odoo.fields import Domain, AND, FALSE_DOMAIN`	`from odoo.fields import Domain` only (`Domain.FALSE`)
`_name_search(..., args=..., name_get_uid=...)`	`(self, name, domain=None, operator='ilike', limit=None, order=None)`
`_read_group_stage_ids(self, stages, domain, order)`	`_read_group_stage_ids(self, stages, domain)`
view modifier `in 'x'` / `not in 'x'`	`== 'x'` / `!= 'x'` (tuple `in ('a', 'b')` is valid)

```
# 20
class AccountAnalyticAccount(models.Model):
    _inherit = 'account.analytic.account'
    _rec_names_search = ('name', 'code', 'custom_number', 'partner_id')
```

```
from odoo.fields import Domain
def _search_display_name(self, *args, **kwargs):
    return Domain(super()._search_display_name(*args, **kwargs))
```

`odoo.tools.Query` is not exported from `odoo.tools` . Import `SQL` from `odoo.tools` .

DOMAIN OPERATOR `access`

20 adds the domain operator `access`. It is valid on `id` and on Many2one fields. The value is `'read'`, `'write'`, `'create'`, or `'unlink'`. Use it when a domain must keep only records (or related records) the user may operate on. It does not work on properties.

```
# 20 — current user can write the partner
Domain('partner_id', 'access', 'write')
# 20 — current user can read this record
Domain('id', 'access', 'read')
```

This operator is useful in `ir.access.csv` domains (Chapter 1) when the rule must follow comodel rights.

4. Binary, Excel, PDF, and time

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CE + EE

BINARY

On the record, a Binary field is `BinaryValue` (`odoo.tools.binary`). `.content` is already raw bytes. `base64.decodebytes(self.datas)` double-decodes or raises `Incorrect padding` .

Direction	Odoo 20
Read	`value.content` or `bytes(value)`
Filename	`value.filename` — optional name on `BinaryValue` / `BinaryBytes`
Write	`from odoo.tools.binary import BinaryBytes` — not raw `bytes` , not `base64.b64encode(...)`
Image write	`image_apply_opt(pil_image, 'PNG')` then `BinaryBytes(...)` . `image_to_base64` is removed
`ir.attachment`	field `raw` , not `datas` ; `res_id` is an integer
XML binary	`type="bytes"`
RPC / HTTP	binary payload is `BinaryValue` , not a bare base64 string

`BinaryBytes(data, filename=name)` keeps the original file name on the value. RPC and HTTP that still send a raw base64 string without that wrapper fail the 20 type check.

```
from odoo.tools.binary import BinaryBytes
from odoo.tools.image import image_apply_opt
record.image_1920 = BinaryBytes(image_apply_opt(pil_image, 'PNG'))
record.datas = BinaryBytes(stream.getvalue(), filename=name)
```

```
<!-- 19 -->
<img t-attf-src="#{doc.image_1920}"/>
<!-- 20 -->
<img t-att-src="image_data_uri(doc.image_1920) if doc.image_1920 else '/web/
static/img/placeholder.png"/>
```

EXCEL

Odoo 19	Odoo 20
`xlwt`	`xlsxwriter` (in the 20 virtualenv). Write to `io.BytesIO` with `{'in_memory': True}` . Assign with `BinaryBytes`
`xlrd.open_workbook` on `.xlsx`	xlrd 2.0.1 cannot open Office Open XML. Use `openpyxl` for `PK` `.xlsx` . Use xlrd only for OLE `.xls`

PDF

Odoo 19	Odoo 20
`import PyPDF2`	`from odoo.tools.pdf import PdfFileWriter, PdfFileReader`
`content_disposition` in `odoo.tools` / `odoo.http` old path	`odoo.http.stream`
`_render_qweb_pdf(template_xmlid)`	`_render_qweb_pdf(action_xmlid)`
`request.env` in a report parser	`env` — `request.env` raises `RuntimeError: request not bound`

TIME

```
from datetime import UTC
from zoneinfo import ZoneInfo
local_tz = ZoneInfo(self.env.user.tz or 'UTC')  # ZoneInfo('utc') fails
if utc_dt.tzinfo is None:
    utc_dt = utc_dt.replace(tzinfo=UTC)
local_dt = utc_dt.astimezone(local_tz)
```

```
resource._attendance_intervals_batch(
    start_dt, end_dt,
    resources_per_tz=resource._get_resources_per_tz(),
)
```

Odoo 19	Odoo 20	
`time_type` on leave / calendar leave	`count_as='absence'`	
`resource.calendar` field `tz`	removed	
calendar attendance `name` / `day_period`	removed	

5. Configuration, HTTP, and mail

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CE + EE unless marked **EE**.

Odoo 19	Odoo 20
`@http.route(..., type='json')`	`@http.route(..., type='jsonrpc')`
`ir.actions.server mail_post_method`	`state="mail_post"` + `template_id` + optional `mail_post_autofollow`
EE server action `use_in_ai`	`ai_tool_name` matching `[a-zA-Z0-9_]{1,64}`
settings inherit `group[@name='preferences']`	`//page[@name='preferences']//group[@name='other_preferences']`
employee settings xpath	after `employees_setting_container`

A mail template that inherits an Enterprise model must use the **defining** module's model xmlid. Example **EE**: `account_budget.model_budget_analytic`. Inherit does not create `your_module.model_budget_analytic`.

`MAIL.TRACKING.VALUE`

CE + EE. 19 stores `mail.tracking.value` in core `mail`. 20 moves the model to the Community addon **`mail_tracking`** (Discuss Tracking). Core `mail` still generates tracking messages; it does not keep the stored tracking rows unless `mail_tracking` is installed.

Odoo 19	Odoo 20
`mail.tracking.value` in `mail`	install `mail_tracking`; then `_name = 'mail.tracking.value'`

Odoo 19	Odoo 20
search `mail.tracking.value` with only `mail`	empty / missing model unless `mail_tracking` is in the graph

Depend `mail_tracking` when custom code reads stored old/new values.

EXTERNAL RPC

`/json/2/<model>/<method>` (JSON-2, `type='json2'`, bearer auth) is the documented external API on 20. The same controller already exists on 19. XML-RPC (`/xmlrpc/2`) is still present on both versions. New integrations should call JSON-2. Do not treat XML-RPC as removed in 20.

```
POST /json/2/res.partner/search_read
```

Controller routes that the web client calls stay `type='jsonrpc'` (table above). That is not the same as JSON-2.

6. SQL reports

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CE + EE

`_table_query` → `_table_sql`

19 SQL views set `_auto = False` and define `_table_query` (attribute or method). 20 **removes** `_table_query`. The lint on 20 is: overwrite `_table_sql`. Return `odoo.tools.SQL`.

```
# 19
_auto = False
def _table_query(self):
    return self._query()
# 20
from odoo.tools import SQL
_auto = False
def _table_sql(self) -> SQL:
    return SQL("(%s)", self._query())
```

`sale.report` on 20 is the reference: `_table_sql()` builds a `Query` and returns `query.subselect(...)`. A module that only migrated `_select()` / `_group_by()` to `SQL` still fails if it leaves `_table_query` in place.

`_select()` AND `_group_by()`

`_select()` and `_group_by()` return `odoo.tools.SQL`. A Python string cannot be concatenated onto that value. `LiteralSQL` has no `.code`.

```
from odoo.tools import SQL
def _select(self):
    return SQL("%s,\n t.custom_type", super()._select())
```

Report	Odoo 20 hook	
`report.project.task.user`	`_select()` / `_group_by()` → `SQL`	
`timesheets.analysis.report`	`_select()` → `SQL`	
`sale.report`	`_select_dict` + `_groupby_list`	
`purchase.report`	`_select_list(table)` + `_groupby_list(table)`	
`account.move` invoice report	`_select_list`	
`report.pos.order`	wrap `_select` only	

A field added on `analytic.view_account_analytic_line_filter` must also be added on `timesheets.analysis.report` through `_select()` .

Translated `name` in SQL:

```
COALESCE(project_task_type.name->>'en_US', project_task_type.name::text)
```

7. Views

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CE + **EE** unless marked otherwise.

KANBAN

Odoo 19 inherit	Odoo 20 inherit
`project.view_project_kanban` body	`project.view_project_card`
`project.view_task_kanban` footer	`project.view_task_card` `//t[@t-name='card']//footer`
kanban template name	`t-name="card"`

`project.view_task_kanban` uses `card_id="%(project.view_task_card)d"` .

RIBBONS AND BUTTON ICONS

```
<!-- 19 -->
<button icon="fa-check"/>
<widget name="web_ribbon" text="Archived" bg_color="bg-info"/>
<!-- 20 -->
<button icon="check"/>
<widget name="web_ribbon" title="Archived" bg_color="text-bg-info"
invisible="active"/>
```

Odoo 19	Odoo 20
`web_ribbon` `text=`	`title=`
`bg_color="bg-info"` / `bg-primary`	`bg_color="text-bg-info"` / `text-bg-primary` / `text-bg-success` / `text-bg-warning` / `text-bg-danger`

Odoo 19	Odoo 20
button `icon="fa-check"`	`icon="check"` — names from `web/icons.py` `ICONS`
Font Awesome / GTK icon names on ViewButton	do not draw. `iconFromString` renders `oi` + `data-icon=`
Font Awesome on website snippets	Material Symbols ligature (same names as ViewButton)

Legal ViewButton names include `check` , `save` , `close` , `history` , `edit_square` , `task` , `group` , `book` , `description` , `stethoscope` .

Owl 3 `DropdownItem` takes `attrs=` . `href=` and `dataset=` are invalid props.

XPATH

View inherit cannot use attribute `string` or `title` as a selector. Use `hasclass(...)` , `name` , `id` , or `t-call` .

Xpath cannot enter the body of a `t-call` . Inherit the called template, or xpath the `t-call` node.

`[@class='X']` is an exact match. Convert the inherit only when the 20 parent class string no longer equals `X` . Use `hasclass('X')` in that case.

Invoice, sale, and purchase order line `product_id` / `name` are inside `<column name="product_and_description">` . `/list/field[@name='product_id']` does not match. Xpath a direct list child (`quantity` , `product_qty` , `analytic_distribution` , `account_id`).

Form	Odoo 19 xpath	Odoo 20
Analytic account	`//header`	no header — insert `<header>` with workflow buttons
Timesheet	add `button_box`	`div[@name='button_box']` already exists
Calendar event	`button_box`	no `button_box` — buttons in `//sheet/widget[@name='web_ribbon']`

Form	Odoo 19 xpath	Odoo 20
Employee	`oe_title`	`//h1` or `//h1[hasclass('o_employee_form_header_info')]`
Job	`user_id`	after `department_id` ; form already uses recruiter_id
CRM	`//div[hasclass('oe_title')]/div`	`//div[hasclass('oe_title')]`
Maintenance request	`archive_equipment_request` / `user_id`	`//header` ; assignees `user_ids`
Invoice totals	`@class='oe_subtotal_footer'`	`//group[hasclass('oe_subtotal_footer')]`
Equipment	19 `location_id`	`owner_user_id`
Partner list	`complete_name`	`display_name`
Partner search	`group[@name='group_by']`	`//search/group`
User preferences	`group[@name='preferences']`	`//page[@name='preferences']`
Product	`product.product_variant_easy_edit_view`	removed
Product action	`product.product_normal_action`	`product.product_variant_action`
Sale config menu	`sale.prod_config_main`	`sale.menu_sale_config`
Project form	`group[@name='extra_settings']`	`//page[@name='settings']`
Timesheet kanban footer	`footer[@class='mt-0 pt-0']`	`div[@name='ts_footer']`

Form	Odoo 19 xpath	Odoo 20
eLearning channel	`//field[@name='active']`	no `active` — xpath `name`
Activity popup	`company_id`	after `summary`
POS session	`move_id`	`user_id`
POS receipt	Owl `ReceiptHeader` / `pos-receipt-contact`	`point_of_sale.pos_order_receipt`
Sale form	`show_update_fpos`	`fiscal_position_id`
Sale portal sidebar	`navspy`	`div#sale_order_sidebar_button`

`product.product_variant_action` context `active_id` is defined only on a product
form. A custom `ir.actions.act_window` copied from that action must not keep
`active_id` if it opens from another model.

8. QWeb and reports

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CE + EE

`<act_window>` and `<report>` are not in Odoo 20 `import_xml.rng`. Use `ir.actions.act_window` and `ir.actions.report` records. `ir.actions.report` has no `report_file`.

```
<record id="action_print_doc" model="ir.actions.report">
    <field name="name">Document</field>
    <field name="model">sale.order</field>
    <field name="report_type">qweb-pdf</field>
    <field name="report_name">my_module.report_document</field>
    <field name="binding_model_id" ref="sale.model_sale_order"/>
    <field name="binding_type">report</field>
</record>
```

Odoo 19	Odoo 20
`t-esc`	`t-out`
`t-raw`	`t-out`
`t-field-options` / `t-esc-options`	`t-options`
`t-field`	still valid
invoice `o.type`	`o.move_type`
BoM overview header `BoM Cost`	`Total Cost`
`from odoo.tools import html2plaintext`	`from odoo.tools.mail import html2plaintext`
`web_editor` Python `get_video_embed_code`	package removed; embed is JS `html_editor`

An xpath that selected `@t-esc` must select `@t-out`.

REPORT ENGINES

CE + EE. 20 splits PDF rendering into two Community modules that 19 did not ship as addons:

Engine	Odoo 20 module	
wkhtmltopdf	`base_report_wkhtmltox`	
Paper Muncher (in-house)	`base_report_paper_muncher`	

A custom report that assumed "one wkhtml pipeline inside `web`" must depend on the engine actually installed. Paper Muncher is the 20 in-house renderer (`https://odoo.github.io/paper-muncher/`).

EE If the report bound to `hr.work.entry` rows: those document records are gone. Bind to `hr.payslip` and iterate `worked_days_line_ids`.

9. Owl 3

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CE + EE

Odoo 20 web is Owl 3.

Odoo 19	Odoo 20
`import { useState } from "@odoo/owl"`	`import { proxy } from "@odoo/owl"` then `this.state = proxy({ ... })`
template `state.foo` / `kpis.foo` / `dashboardData.foo`	`this.state.foo` / `this.kpis.foo` / `this.dashboardData.foo` / `this.props.foo`
`t-esc`	`t-out`
`t-model` on a `proxy()` field	`t-model` accepts a `signal` only. `proxy()` fields use `t-att-value` + `t-on-input`
`t-set` inside `t-call` as a sub-template variable	slot body — pass parameters as attributes on `<t t-call="..."/>`
`standardActionServiceProps` from `action_service`	`@web/webclient/actions/action_plugin`
Chart.js v2 options	Chart.js v4 (`cutout` , `plugins.legend` , `plugins.title` , `scales.y.beginAtZero`) at `web/static/lib/Chart/Chart.js`
`_attachment_format`	`_store_attachment_fields`
`mail.AttachmentList` props `attachments`	`attachmentGroups` ; container `o-mail-AttachmentContainer`
`mail.assets_messaging`	empty — put JS/XML in `web.assets_backend`
`window.alert` in systray	notifications; company is `user.activeCompany`

```
import { proxy } from "@odoo/owl";
this.state = proxy({ total: 0 });
```

```
<t t-out="this.state.total"/>
<t t-out="this.kpis.open"/>
```

`t-as` loop locals stay unprefixed. A static CSS class is not a template identifier (`ud-widget` stays `ud-widget`).

```
static props = { "*": true };
```

```
import { signal } from "@odoo/owl";
this.searchTerm = signal("");
```

Mail store **CE + EE**:

```
Store().add(channel, "_store_channel_fields")._build_result()
```

```
this.store["mail.thread"].getOrFetch({ model: "discuss.channel", id })
```

A Python method called from Owl as `[[], payload]` must be `@api.model` .

Owl 3 also drops or relocates several 19 hooks. Replace them when the 20 component fails to compile or never re-renders.

Odoo 19	Odoo 20
`this.render()`	the `proxy` / `signal` update re-renders; do not call `render`
`t-portal`	removed — render into the target node, or a 20 portal helper if the standard component still exposes one
`onWillUpdateProps` / `onWillRender` / `onRendered`	removed from Owl 3. Use `onWillStart` , `onMounted` , `onWillUnmount` , and `onPatched`

10. Accounting and payments

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CE + EE unless marked EE.

Odoo 19	Odoo 20
invoice lines `journal_line_ids`	`line_ids`
invoice salesperson group	`sale_info_group` (customer invoices)
vendor bill extra fields	`accounting_info_group`
journal entry other info	`other_tab_entry`
header dates / flags	`header_right_group`
`res.partner.bank.acc_number`	`account_number`
`res.partner.bank.acc_holder_name`	`holder_name`
`res.partner.bank.bank_id.name` / `bank_id.bic`	`bank_name` / `bank_bic`
`account.journal.bank_id`	`bank_name`
`res.bank`	removed — Char, or `res.partner.bank`
`account.analytic.account.group_id`	`plan_id`
`payment.provider.state`	`is_live` / `is_published`
`payment.controllers.post_processing`	removed
payment transaction group `//group//group`	`//group[@name='transaction_details']`
invoice report `o.type`	`o.move_type`
`account.group`	removed
`account.account.group_id`	`parent_id` / `parent_ids` on `account.account`

Odoo 19	Odoo 20
account `code` required	`code` is computed / optional (`code_store` , `placeholder_code`)
`account.payment` state `in_process`	`paid`
`account.payment` state `paid`	`reconciled`

ACCOUNT GROUPS → PARENT ACCOUNTS

19 has `account.group` and `account.account.group_id` . 20 deletes `account.group` . Hierarchy lives on `account.account` :

```
# 20
parent_id = fields.Many2one('account.account', string="Parent Account")
parent_ids = fields.Many2many('account.account',
compute='_compute_parent_ids')
```

Replace searches on `account.group` and writes to `group_id` with parent accounts. Account codes are no longer a required stored Char; 20 computes `code` (`compute_sql`). Reports that `JOIN` a mandatory `account_account.code` must tolerate empty codes and use `placeholder_code` when displaying.

PAYMENT STATES

Odoo 19	Odoo 20	
`draft`	`draft`	
`in_process`	`paid`	
`paid`	`reconciled`	
`canceled`	`canceled`	

Filters, automations, and domains that still use `in_process` match nothing. A payment that 19 called "Paid" is **Reconciled** on 20. A payment that 19 called "In Process" is **Paid** on 20.

FIXED ASSETS — EE

19 Enterprise treats a depreciation template as `account.asset` with `state == 'model'` . 20 Enterprise adds **`account.depreciation.model`** . Bind templates, default depreciation methods, and CoA-generated models to that model. `account.asset` remains the asset document.

`payment.method.provider_id` is required. Create the provider first. There is no `payment.payment_method_card` xmlid in `payment` .

Custom fields copied from payment to the journal entry go in `_generate_move_vals` / `_synchronize_to_moves` .

Invoice line `_compute_account_id` assigns `line.account_id` . It does not `return` .

11. Sales, purchase, and products

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CE + EE

Odoo 19	Odoo 20
`sale.order` state `done`	removed. Valid: `draft` / `sent` / `sale` / `cancel`
`purchase.order.line` UoM `product_uom_id`	`uom_id`
`sale.order.line` / `account.move.line` UoM	still `product_uom_id`
`product.type == 'product'`	`is_storable`
`product.pricelist.item._compute_price(..., date)` positional	`date=` keyword (`_compute_base_price` the same)
`sale.prod_config_main`	`sale.menu_sale_config`
`product.product_normal_action`	`product.product_variant_action`
`product.product_variant_easy_edit_view`	removed

`sale.order.line` still has `product_uom_qty` and `product_uom_id`.

12. Inventory, MRP, and repair

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CE + EE

Odoo 19	Odoo 20
`stock.scrap`	`stock.move` with `is_scrap=True`
scrap `name`	`reference`
scrap / move UoM `product_uom` / `product_uom_id`	`uom_id`
scrap dest `scrap_location`	`company.scrap_location_id`
scrap validate	`action_scrap()`
scrap form / action	`stock.view_scrap_move_form` / `stock.stock_scrapped_moves_action`
`stock.return.picking`	`stock.picking._create_return()`
`stock.quant.package`	`stock.package`
`stock_picking_batch`	merged into `stock` ; inherit `stock.stock_picking_batch_form`
`mrp.bom` / `mrp.bom.line` / `repair.order` UoM	`uom_id`
picking move list UoM	`//field[@name='move_ids']/list/field[@name='uom_id']`
`mrp.production.show_produce_all`	removed
MO button “Produce All” (separate from Produce)	one `button_mark_done`

19 manufacturing orders expose `show_produce_all` and two header buttons (Produce vs Produce All). 20 removes that field. Close the order with `button_mark_done` only. Xpaths and automations that test `show_produce_all` fail.

13. HR, time off, and payroll

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CE + EE for employee / time off. **EE** for Payroll.

Odoo 19	Odoo 20
`hr.contract`	`hr.version`
SQL join `hr_contract`	`hr_version`
`hr.version.wizard`	`employee.contract_template_id` + `_onchange_contract_template_id()`
`hr.leave.type`	`hr.work.entry.type`
`hr.leave.holiday_status_id`	`work_entry_type_id`
`hr.leave.action_confirm`	removed; `action_approve` remains
`hr.leave.holiday_type`	`validation_type`
inherit time-off type form	`hr_holidays.hr_work_entry_type_view_form` `support_document`
`hr.expense.sheet`	`hr.expense`
`hr_work_entry.work_entry_type_attendance`	`hr_work_entry.generic_work_entry_type_attendance`
legal leave default	`hr_work_entry.generic_work_entry_type_legal_leave`
leave XML on `hr.leave.type`	merge onto `hr.work.entry.type` (`count_as='absence'` , `time_off_selectable`
depend `hr_work_entry_holidays` only	`hr_holidays`
EE `hr.payslip.input.type`	removed; `input_usage_payslip` on the salary structure

Odoo 19	Odoo 20
EE salary rule `category_id` / `struct_id`	`category_ids` / `struct_ids` `[( 6 , 0, [ref(...)])]`
EE `payslip.paid_amount`	`payslip._get_basic_salary()`
EE payment wizard `export_format` `selection_add`	`_get_export_format_selection()`
EE pay-run kanban	`hr_payroll.hr_payrun_payslip_kanban_card`
EE `hr.work.entry` list documents	removed; print from `hr.payslip` `worked_days_line_ids`

Search filter `approve` on time off is nested. `//search/filter[@name='approve']` does not match. Use `//filter[@name='approve']` .
Several 20 workflows use state `canceled` , not `cancel` .

14. Project, timesheets, and portal

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CE + EE unless marked EE.

Odoo 19	Odoo 20
`project.task.user_id`	`user_ids`
access domain `task_id.user_id`	`('task_id.user_ids', 'in', [user.id])`
`note.note`	`project.task` (`project_todo`)
todo `default_memo`	`default_name` (`project.task` has no `memo`)
`project.portal_my_home`	`portal.entry` (`project.portal_projects` , `project.portal_tasks`)
portal card `active`	`show_in_portal`
`portal_client_category` / `portal_service_category`	removed
`portal.portal_sidebar` `//t[@t-set='entries']`	`//div[hasclass('col-lg-3')]`
invoice portal download	`//div[hasclass('o_download_pdf')]`
portal docs wrap	`div.o_portal_docs`
`get_records_pager`	removed

`_prepare_home_portal_values` counters are merged in `counters()` . Group-restricted cards use `is_config_card` and `_filter_visible_portal_cards` .

PROJECT PROFITABILITY — CE

19 Community `project.project` implements `_get_profitability_items` , `_get_profitability_values` , `_get_profitability_labels` , and related helpers. Those

methods are **not** on Community 20 `project`. Overrides and tests that call them raise `AttributeError`. Enterprise 20 reports margin and budget through the budget apps (Chapter 17), not through those 19 CE hooks.

EE Field Service: `industry_fsm` is not in Enterprise 20. Depend `project` (and `sign` if signatures are used).

Odoo 19 EE	Odoo 20	
`industry_fsm`	`project`	
`industry_fsm.fsm_menu_root`	`project.menu_main_pm`	
`industry_fsm.group_fsm_user`	`project.group_project_user`	

`is_fsm` is not a standard 20 field. Custom domains that still use it need a local Boolean on `project.project` / `project.task`. Enterprise Planning (`planning.slot`) is a different model.

15. CRM, calendar, and mail

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CE + EE

Odoo 19	Odoo 20
CRM `//div[hasclass('oe_title')]/div`	`//div[hasclass('oe_title')]`
lead `currency_id` (standard)	`company_currency`
notebook `page[@name='lead']`	`internal_notes` / `extra`
Extra Info `company_id` in `categorization`	`//page[@name='extra']//field[@name='company_id']` (`group[@name='Misc']`)
graph amount	`type="measure"`
calendar `button_box`	removed — see Chapter 7
calendar `page_invitations`	`notes` / `options`
`mail.tracking.value` in `mail`	Community addon `mail_tracking` (Chapter 5)

16. Website and eCommerce

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CE + EE

```
<!-- 19 — 500 on 20 public pages -->
<input t-attf-value="#{user_id.partner_id.email}"/>
<!-- 20 -->
<input t-att-value="request.env.user.partner_id.email if not
request.env.user._is_public() else ''"/>
```

Odoo 19	Odoo 20
`a[@id='add_to_cart']`	`button[@name='add_to_cart']`
add-to-cart wrap	`div[@id='add_to_cart_wrap']` / `website_sale.cta_wrapper` `#o_wsale_cta_wrapper`
`product._is_add_to_cart_possible()` form	removed
`website_sale.product_terms_and_conditions`	removed — xpath after `#o_wsale_product_cta_section`
`#wrap.js_sale`	removed
`website_sale_wishlist` / `website_sale_comparison`	merged into `website_sale`
wishlist / compare xmlids	`website_sale.product_wishlist` , `website_sale.product_compare` `website_sale.model_product_wishlist`
comparison controller	`odoo.addons.website_sale.controllers.cart`
hide price	`combination_info['hide_price']`
category form `sequence`	list only — xpath `parent_id`
`res.partner.company_type`	removed — `is_company`

Odoo 19	Odoo 20
`res.partner.mobile`	removed — `phone`
`res.users.groups_id`	`group_ids`
action context `default_company_type`	`default_is_company`
`ir.ui.view.groups_id`	unchanged (view access, not the user field)
Font Awesome class on website snippets (`fa` `fa-...`)	Material Symbols ligature / `oi` + `data-icon` rendering
`request.website` in a model method	`self.env.website` (Chapter 2)

Website and snippet markup that still ships Font Awesome class names will not draw on 20. Use the same ViewButton / Material Symbols names as Chapter 7 (`check` , `close` , …). Public pages that read `request` from a model or report parser hit `request not bound` ; use `env.website` .

```
# 19
vals['company_type'] = 'person'
vals['mobile'] = '...'
user_vals['groups_id'] = [(6, 0, ids)]
# 20
vals['is_company'] = False
vals['phone'] = '...'
user_vals['group_ids'] = [(6, 0, ids)]
```

17. Enterprise budget, Field Service, Helpdesk, and Fleet

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BUDGET — EE

Odoo 19 Enterprise	Odoo 20 Enterprise	
`crossovered.budget`	`budget.analytic`	
`crossovered.budget.lines`	`budget.line`	
lines field	`budget_line_ids`	
amount field	`budget_amount`	
states	`draft` / `confirmed` / `revised` / `done` / `canceled`	
form	`account_budget.view_budget_analytic_form`	
action	`account_budget.act_budget_analytic_view`	
report binding	`account_budget.model_budget_analytic`	
`committed_amount`	on `account_budget_purchase`	

CE Community 20 has no `budget.analytic` . A Community budget module that still uses its own model is not the Enterprise model and is not retargeted to `account_budget` .

HELPDESK

EE Enterprise `helpdesk` remains. CE a custom `helpdesk.sla` `_name` must not `_inherit = 'helpdesk.sla'` unless Enterprise Helpdesk is in the registry (Chapter 2).

These 19 Enterprise addons have **no** 20 folder. Depend `project` / `helpdesk` / `documents` instead of the FSM bridges:

Odoo 19 Enterprise	Odoo 20	
`industry_fsm*`	`project` (Chapter 14)	
`helpdesk_fsm` / `helpdesk_fsm_report` / `helpdesk_fsm_sale`	`helpdesk`	
`documents_fsm`	`documents`	

FIXED ASSETS — EE

See Chapter 10. 20 adds `account.depreciation.model` . 19 templates were `account.asset` with `state == 'model'` .

AI ADDONS — EE

Enterprise 20 ships a set of `ai_*` modules. They are new optional apps, not a rename of a 19 module. Do not add them to a 19 graph unless the 20 database actually uses that stack.

FLEET — CE + EE

Odoo 19	Odoo 20	
`fleet.vehicle.log.services.date`	`date_from` / `date_to`	

18. Symptom index

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Exception or result	Chapter	
`KeyError: 'ir.rule'`	1	
`Field has no SQL representation`	1	
`use BinaryValue instead of bytes`	4	
`Incorrect padding` / invalid `.xlsx`	4	
`No module named xlwt` / `pytz`	4	
`stock.scrap` missing	12	
`hr.leave.type` / `hr.contract` missing	13	
`hr.expense.sheet` missing	13	
`industry_fsm` missing	14, 17	
`crossovered.budget` missing	17	
`Invalid field 'company_type'` / `'groups_id'` / `'mobile'`	16	
`mail.thread.cc` / MRO error	2	
`useState is not a function`	9	
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`LiteralSQL` / `_select` concatenate	6	
xpath parse / does not match	7	
`<act_window>` / `<report>` RNG	8	
sale `done` never matches	11	
portal `/my` cards missing	14	

Exception or result	Chapter	
`get_param` / `set_param` missing	2, 5	
`check_access_rights` / `_name_search` missing	2, 3	
`request not bound`	4	
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ViewButton icon blank	7	
`_table_query` / overwrite `_table_sql`	6	
`mail.tracking.value` missing	5, 15	
`account.group` missing	10	
payment `in_process` never matches	10	
`show_produce_all` missing	12	
`_get_profitability_items` missing	14	
`helpdesk_fsm` / `documents_fsm` missing	17	
`account.depreciation.model` missing on 19 code	10, 17	
Font Awesome icon blank on website	7, 16	
`request.website` / `env.website`	2, 16	

19. Mapping tables

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MODELS

Odoo 19	Odoo 20	Edition
`hr.contract`	`hr.version`	CE + EE
`hr.leave.type`	`hr.work.entry.type`	CE + EE
`hr.expense.sheet`	`hr.expense`	CE + EE
`hr.payslip.input.type`	removed	EE
`stock.scrap`	`stock.move` (`is_scrap=True`)	CE + EE
`stock.return.picking`	`stock.picking._create_return()`	CE + EE
`stock.quant.package`	`stock.package`	CE + EE
`note.note`	`project.task`	CE + EE
`mail.thread.cc`	`mail.thread`	CE + EE
`res.bank`	Char or `res.partner.bank`	CE + EE
`industry_fsm`	`project`	EE
`stock_picking_batch`	`stock`	CE + EE
`website_sale_wishlist`	`website_sale`	CE + EE

Odoo 19	Odoo 20	Edition
website_sale_comparison	website_sale	CE + EE
crossovered.budget	budget.analytic	EE
crossovered.budget.lines	budget.line	EE
hr.version.wizard	removed	CE + EE
account.group	account.account (parent_id)	CE + EE
account.asset state='model'	account.depreciation.model	EE
mail.tracking.value (in mail)	mail.tracking.value (in mail_tracking)	CE + EE
helpdesk_fsm / helpdesk_fsm_report / helpdesk_fsm_sale	helpdesk	EE
documents_fsm	documents	EE

FIELDS

Model	Odoo 19	Odoo 20	Edition
sale.order	state done	removed	CE + EE
stock.move.line / stock.move / purchase.order.line / mrp.bom / repair.order	product_uom_id / product_uom	uom_id	CE + EE
sale.order.line / account.move.line	product_uom_id	unchanged	CE + EE
res.partner.bank	acc_number , acc_holder_name	account_number , holder_name	CE + EE
res.partner	company_type , mobile	is_company , phone	CE + EE

Model	Odoo 19	Odoo 20	Edition
`res.users`	`groups_id`	`group_ids`	CE + EE
`ir.attachment`	`datas`	`raw`	CE + EE
`project.task`	`user_id`	`user_ids`	CE + EE
`product.template`	`type == 'product'`	`is_storable`	CE + EE
`hr.leave`	`holiday_status_id`	`work_entry_type_id`	CE + EE
`hr.leave`	`holiday_type`	`validation_type`	CE + EE
`account.analytic.account`	`group_id`	`plan_id`	CE + EE
`crm.lead`	`currency_id`	`company_currency`	CE + EE
`fleet.vehicle.log.services`	`date`	`date_from` / `date_to`	CE + EE
`payment.provider`	`state`	`is_live` / `is_published`	CE + EE
`resource.calendar.leaves`	`time_type`	`count_as`	CE + EE
`account.move` report	`type`	`move_type`	CE + EE
`account.account`	`group_id`	`parent_id` / `parent_ids`	CE + EE
`account.payment`	state `in_process`	`paid`	CE + EE
`account.payment`	state `paid`	`reconciled`	CE + EE

Model	Odoo 19	Odoo 20	Edition
`mrp.production`	`show_produce_all`	removed (`button_mark_done`)	CE + EE
Binary	bytes / base64	`BinaryValue` + `.filename`	CE + EE

XMLIDS AND VIEWS

Odoo 19	Odoo 20
`sale.prod_config_main`	`sale.menu_sale_config`
`product.product_normal_action`	`product.product_variant_action`
`product.product_variant_easy_edit_view`	removed
`project.view_project_kanban` body	`project.view_project_card`
`project.view_task_kanban` card	`project.view_task_card`
`project.portal_my_home`	`portal.entry`
`industry_fsm.fsm_menu_root`	`project.menu_main_pm`
`hr_work_entry.work_entry_type_attendance`	`generic_work_entry_type_attendance`
pay-run kanban	`hr_payroll.hr_payrun_payslip_kanban_card`
Enterprise budget form	`account_budget.view_budget_analytic_form`
scrap form	`stock.view_scrap_move_form`
batch picking form	`stock.stock_picking_batch_form` in `stock`
`_table_query`	`_table_sql()` → `SQL`
`request.website` in models	`env.website`
wkhtml inside `web`	`base_report_wkhtmltox` / `base_report_paper_muncher`

Odoo 19	Odoo 20
JSON-2 `/json/2`	still present (preferred external API); XML-RPC / xmlrpc/2 still present

Odoo Migration Cookbook: Version 19 to Version 20

Community Edition and Enterprise Edition First Edition, September 2026

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